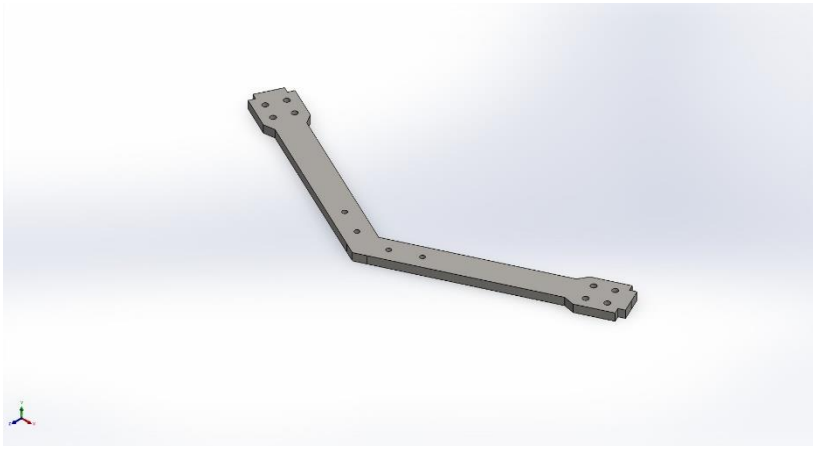


Simulation of V bracket (V)

Date: Wednesday, December 11, 2024
Designer: Solidworks
Study name: Stress Analysis
Analysis type: Static

Table of Contents

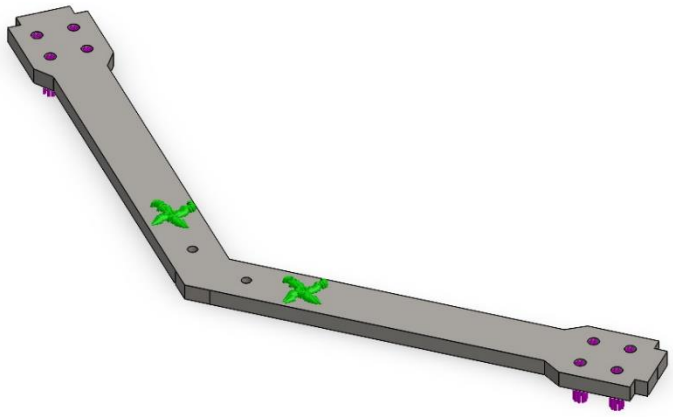
Description 1
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Description

No Data

Assumptions

Model Information



Model name: V bracket (V)
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Cut-Extrude2 	Solid Body	**Mass:0.191255 kg** **Volume:2.45198e-05 m^3** **Density:7,800 kg/m^3** **Weight:1.87429 N**	**C:\Users\TD\Desktop\Final CAD Project\Our Project\Our Project\Saeed Parts\V bracket (V).SLDPRT** Dec 11 23:39:13 2024

Study Properties

Study name	Stress Analysis
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\TD\Desktop\Final CAD Project\Our Project\Our Project\Saeed Parts)

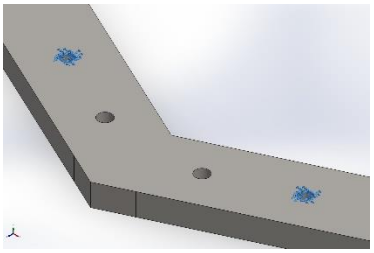
Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m^2

Material Properties

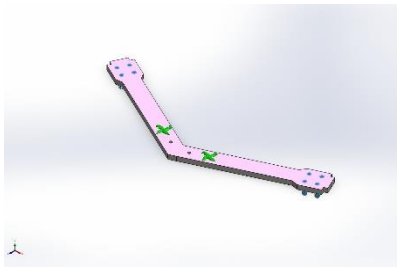
Model Reference	Properties		Components
	**Name:** **Model type:** **Default failure criterion:** **Yield strength:** **Tensile strength:** **Elastic modulus:** **Poisson's ratio:** **Mass density:** **Shear modulus:** **Thermal expansion coefficient:**	Cast Carbon Steel Linear Elastic Isotropic Max von Mises Stress 2.48168e+08 N/m^2 4.82549e+08 N/m^2 2e+11 N/m^2 0.32 7,800 kg/m^3 7.6e+10 N/m^2 1.2e-05 /Kelvin	SolidBody 1(Cut-Extrude2)(V bracket (V))
Curve Data:N/A			

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		**Entities:** 2 edge(s) **Type:** Fixed Geometry

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	1.19209e-06	-6	3.33786e-06	6
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		**Entities:** 8 edge(s) **Reference:** Face< 1 > **Type:** Apply force **Values:** ---, ---, 0.75 N

Connector Definitions

No Data

Interaction Information

No Data

Mesh information

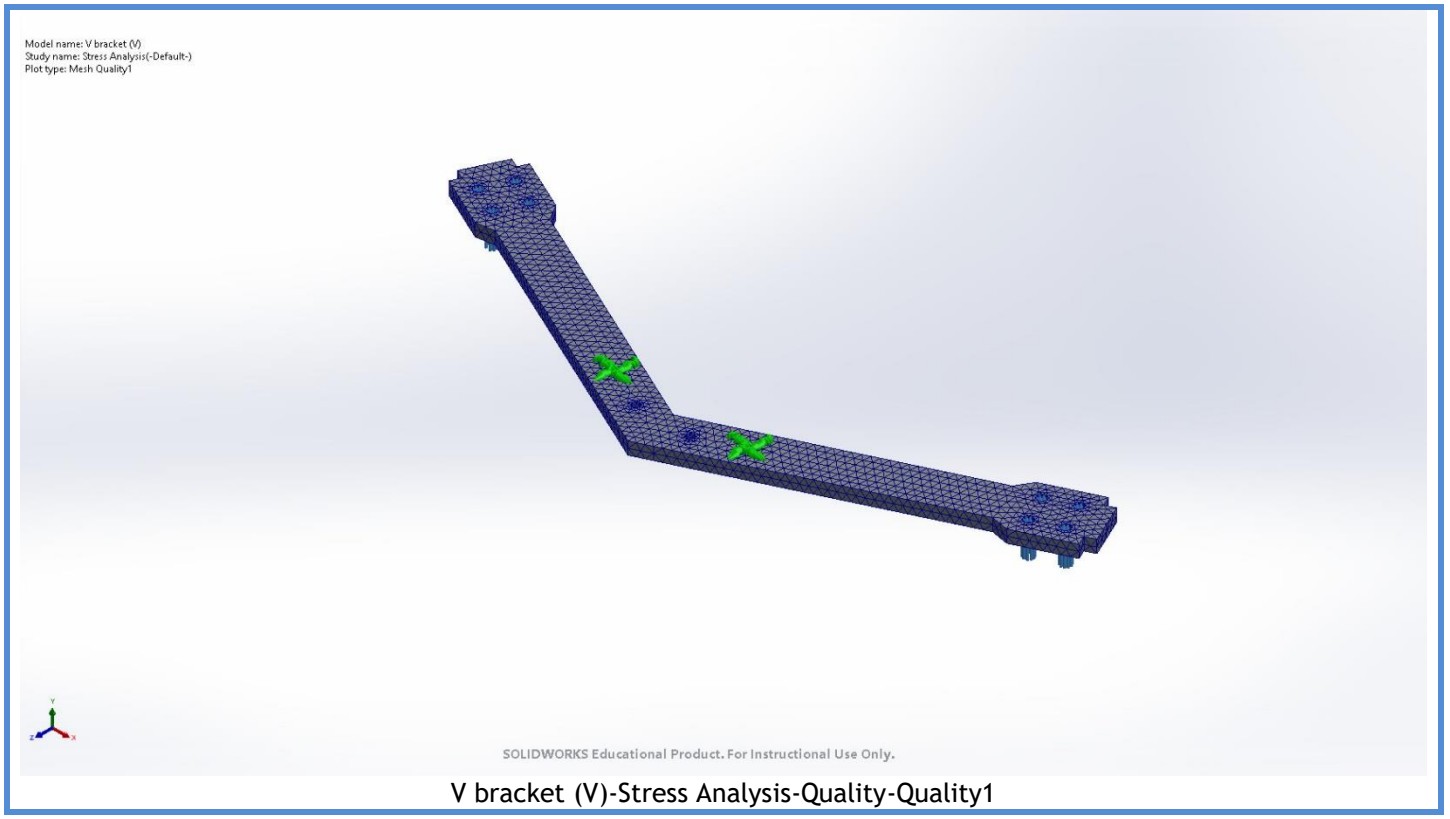
Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	2.9062 mm
Minimum element size	0.968725 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	21268
Total Elements	12113
Maximum Aspect Ratio	7.5588
% of elements with Aspect Ratio < 3	97.7
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh;mm;ss):	00:00:04
Computer name:	

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



V bracket (V)-Stress Analysis-Quality-Quality1

Sensor Details

No Data

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	1.19209e-06	-6	3.33786e-06	6

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	1.19209e-06	1.21593e-05	1.43051e-06	1.23011e-05

Free body moments

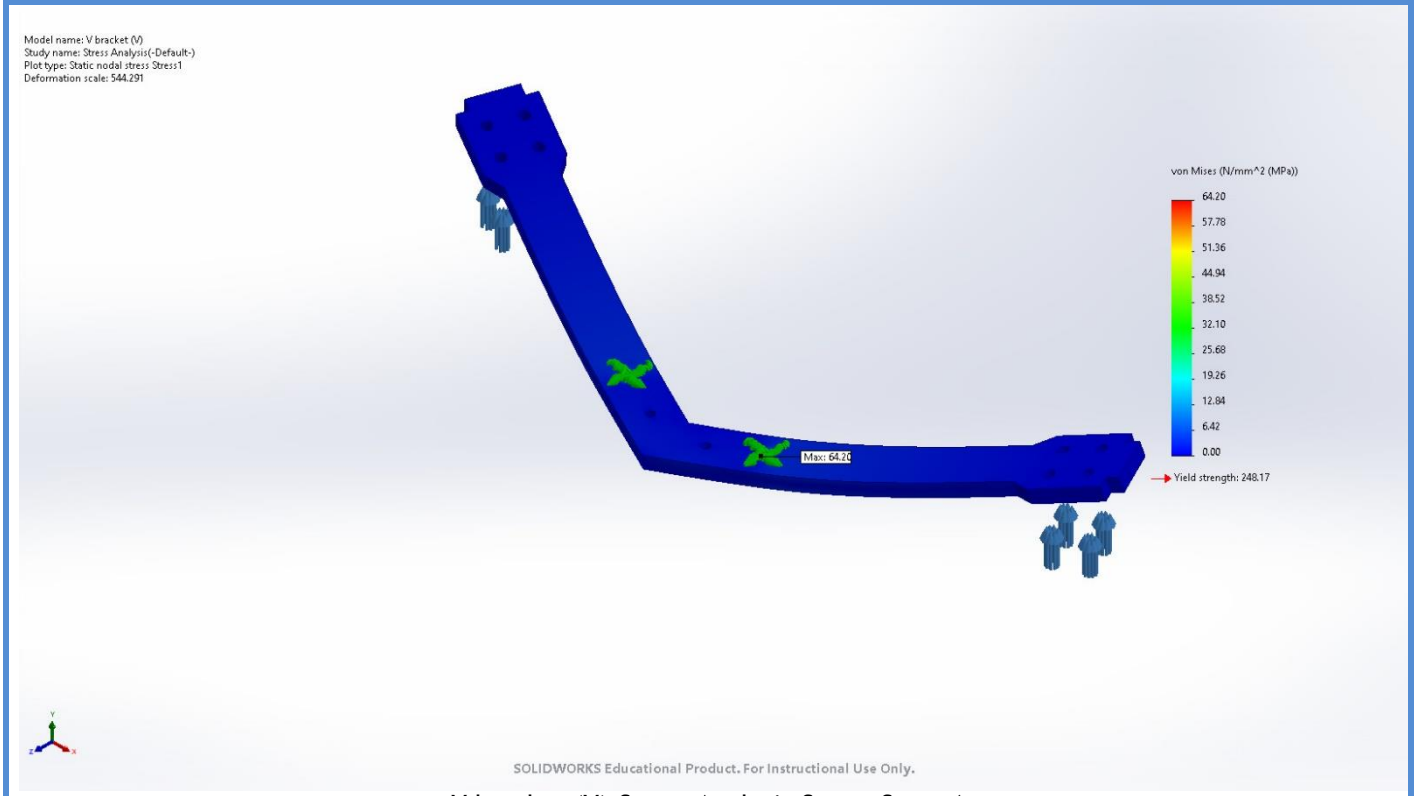
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data

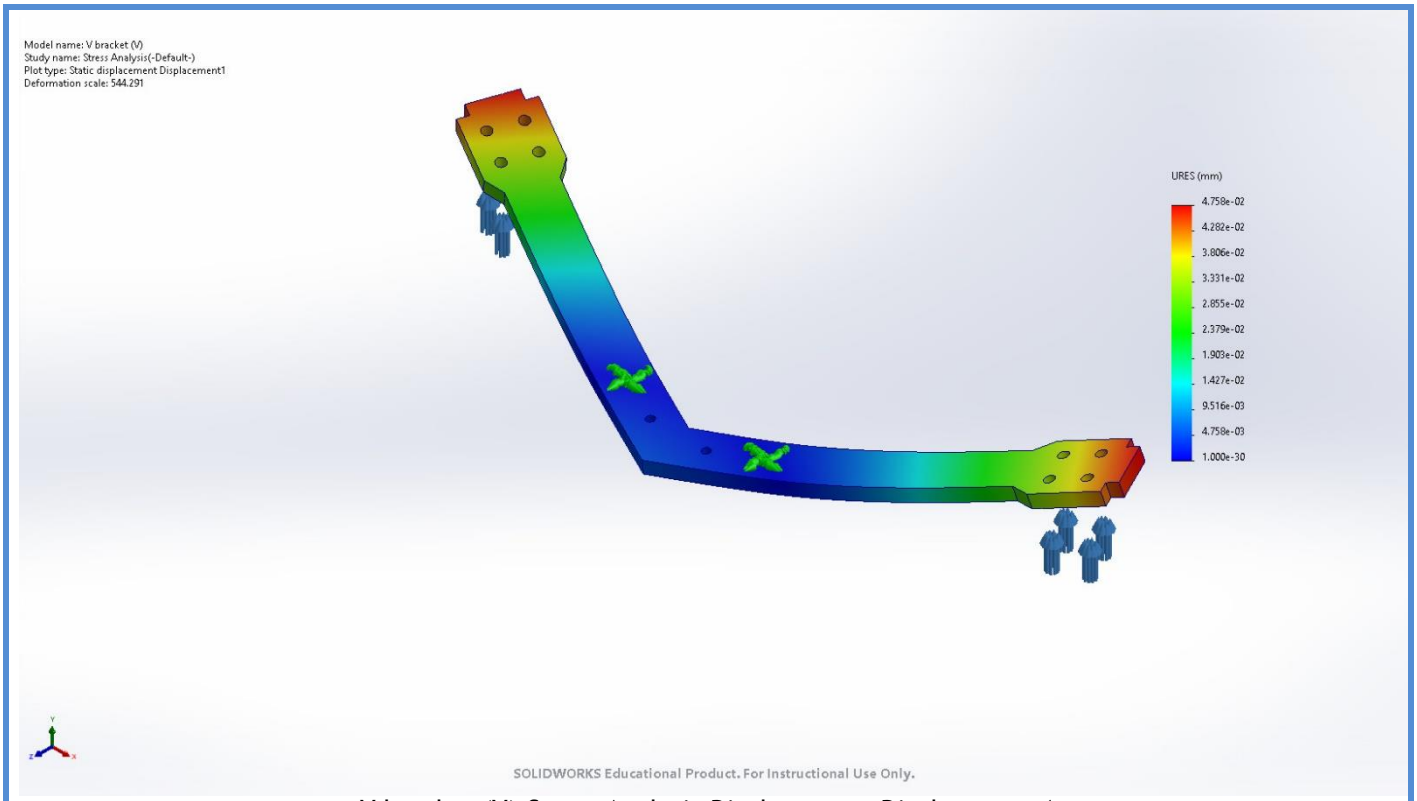
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	0.00N/mm^2 (MPa) Node: 16535	64.20N/mm^2 (MPa) Node: 20145



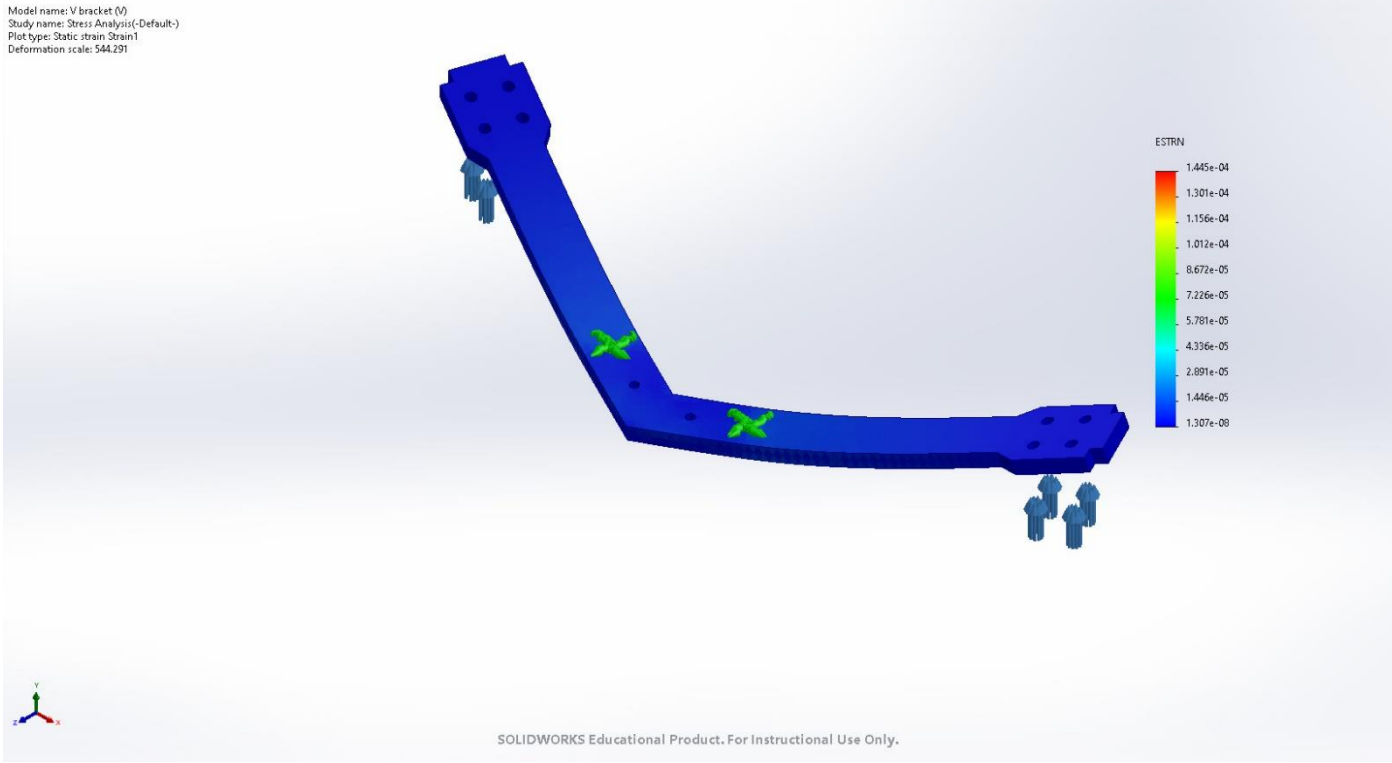
V bracket (V)-Stress Analysis-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 69	4.758e-02mm Node: 17



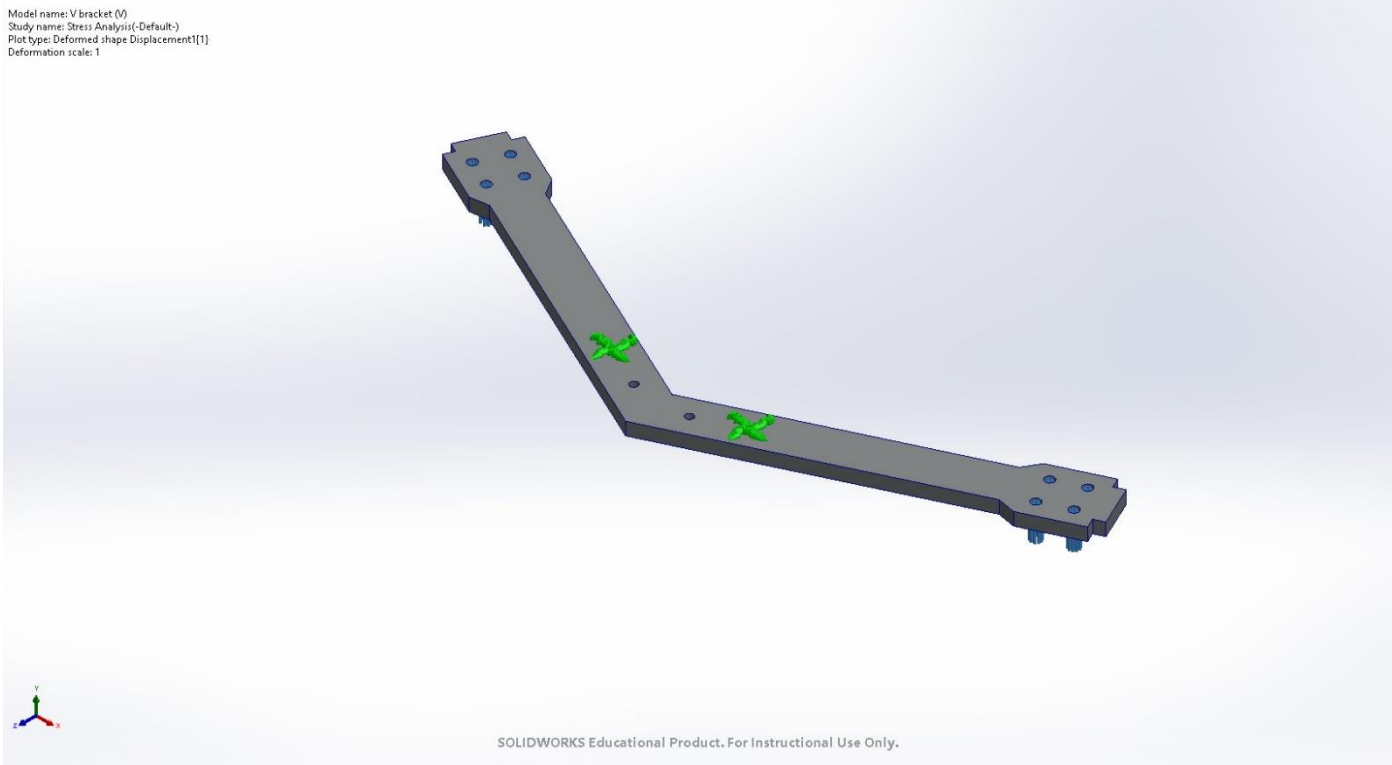
V bracket (V)-Stress Analysis-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	1.307e-08 Element: 9410	1.445e-04 Element: 8387



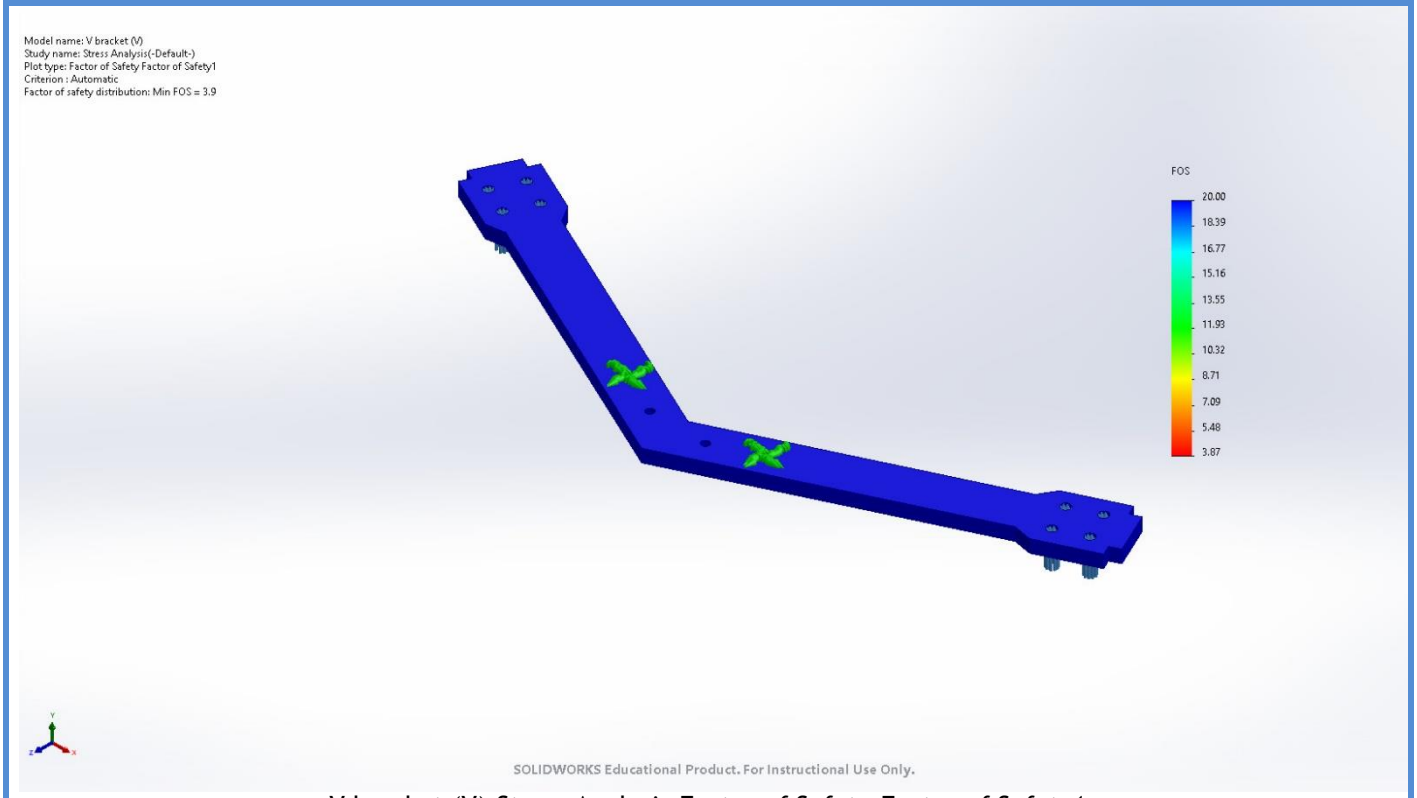
V bracket (V)-Stress Analysis-Strain-Strain1

Name	Type
Displacement1{1}	Deformed shape



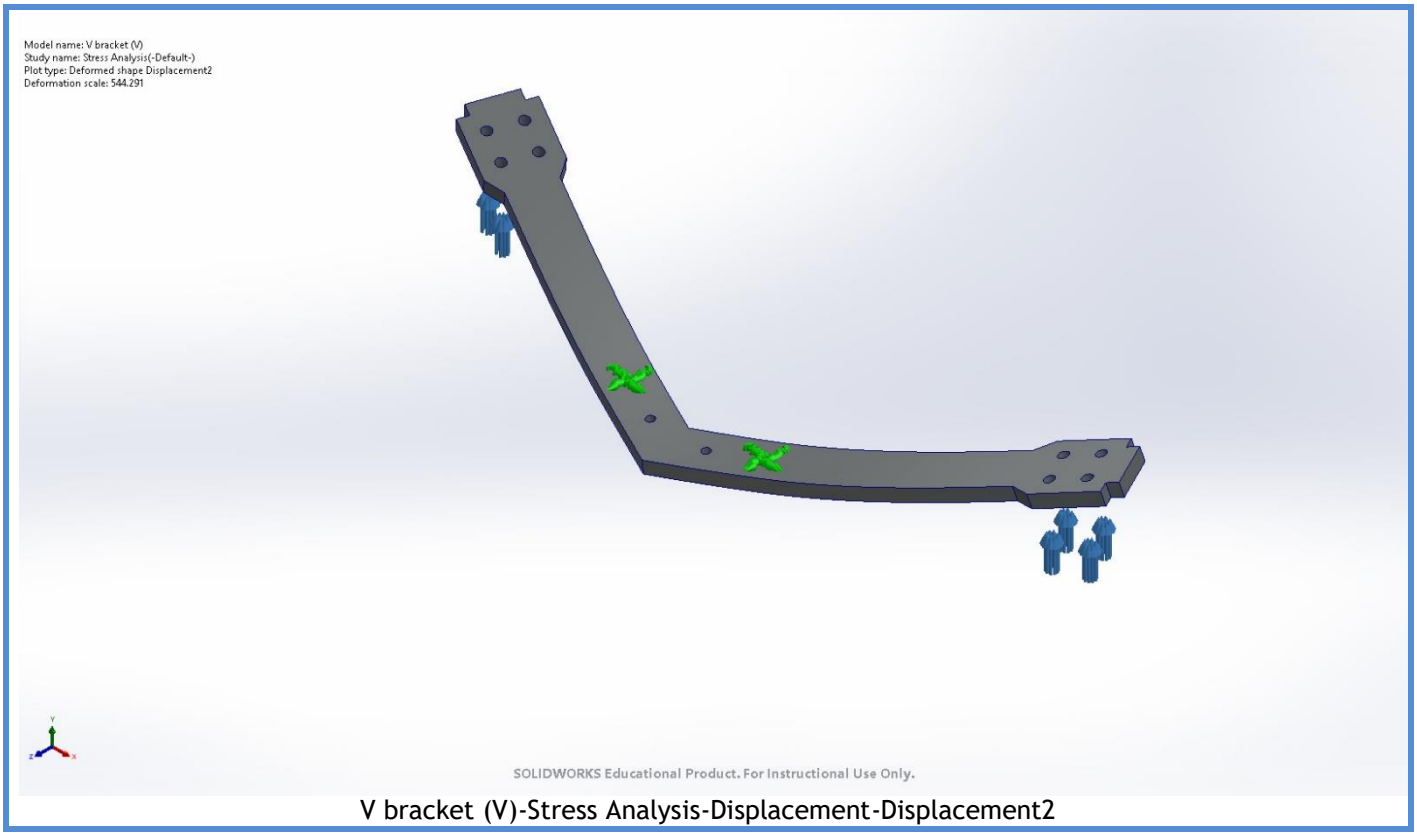
V bracket (V)-Stress Analysis-Displacement-Displacement1{1}

Name	Type	Min	Max
Factor of Safety1	Automatic	3.87 Node: 20145	1,693,870.63 Node: 16535



V bracket (V)-Stress Analysis-Factor of Safety-Factor of Safety1

Name	Type
Displacement2	Deformed shape



V bracket (V)-Stress Analysis-Displacement-Displacement2

Conclusion